

CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV) and the tracking of volatility** regime transitions, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

This paper's honest thesis. We pre-registered a falsifiable advantage hypothesis (H0) and then *attacked our own result* with the strongest fair controls. The finding is a **carefully bounded negative: at simulable scale (≤ 16 qubits) the quantum reservoir shows no statistically significant forecasting advantage** over strong classical baselines once they are given the same information; the dominant lever is *which features are encoded*, not the reservoir's quantum nonlinearity. We report what *does* survive — measurable kernel distinctness, lower seed variance, and full-rank entanglement (classical-simulation hardness) — as quantum-specific *properties* that are necessary, not sufficient, for advantage.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded $R_Y(\pi \cdot x)$, one value per qubit, onto $|0\dots 0\rangle$; the state evolves under a fixed transverse-field **Ising** Hamiltonian $H = \sum_{\{i<j\}} J_{ij} Z_i Z_j + h_x \sum_i X_i$ ($h_x=1$; J a **random $\approx 50\%$ -connected** graph, connectivity=0.5; $U=\exp(-iH\tau)$, $\tau=2$); single- and pairwise Pauli-Z expectations $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$ form the feature vector. A **ridge head** consumes these features concatenated with a linear block of the same inputs (incl. HAR), so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale τ -bank; RZ measurement feedback; regime-adaptive Ising $\leftrightarrow$ Heisenberg via BOCPD; dissipation-as-feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to $\approx 5 \times 10^{-1}$ and compiles, at $n=8$, to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the $\approx 50\%$ -connected graph (an all-to-all reservoir would need ≈ 720).

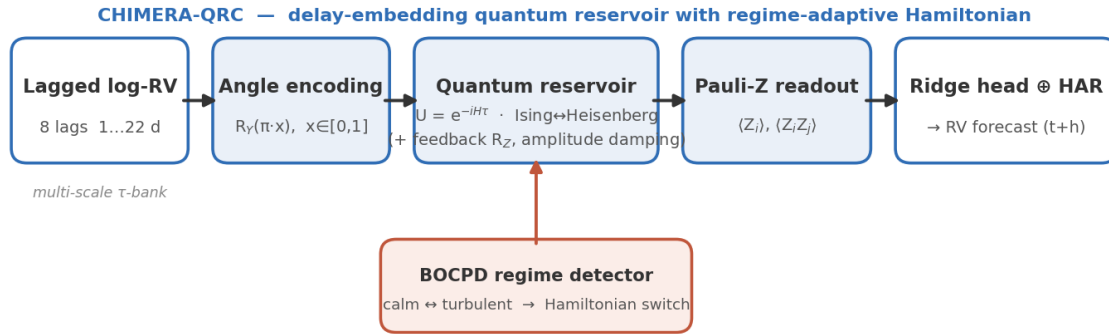


Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** — n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost; (b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **fading memory** (echo-state property); (d) **Hamiltonian as inductive bias**. The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: against the name-matched ESN-108 reference, **$g(\text{ESN} \rightarrow \text{CHIMERA}) \approx 62$ vs ≈ 4 control** (`cli.py run kernel`); against per- n *feature-matched* ESNs the gap is larger still ($g \approx 125$ at $n=8$, rising to ≈ 158 at $n=10$). (g is reported at a fixed ridge regularization; the qualitative $\sim 15\text{--}40\times$ separation over the classical-classical control, not the exact value, is the claim.) Crucially, at $\leq 12\text{--}16$ qubits the map remains classically simulable, so distinctness is **necessary, not sufficient** for accuracy — and §5.3 shows that, here, it is indeed *not* sufficient.

4. Data modeling strategy

Datasets (all public). Oxford-Man .SPX 5-minute realized variance, 2000–2020, 5,029 supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`); public SPY daily OHLCV 2022–2026 (Garman–Klass proxy); real MNIST (Keras .npz mirror). **Preprocessing:** log-RV target; features min-max scaled to $[0,1]$ on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥ 1 day; train-only scaling). **Baselines:** HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + $4\times$, recurrent), RFF kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2 ; significance by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, extra qubits receive no new input: $g(n)$ and effective feature-rank

saturate/decline (n=8→12: g 133→30, D_eff 1.8→1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at n=10, informed vs idle qubits, kernel distinctness and rank jump (**g 52→158, D_eff 1.5→3.1**), and effective rank grows with n (D_eff 1.81→3.10→3.14, n=8→12; near-flat 10→12). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

5.3 The decisive, adversarially-controlled test (the honest headline). We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds, HAC-DM, two windows, and Holm correction** (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis n=8	0.6255	0.6386	0.6302	0.6379	HAR-X
crisis n=10	0.6034	0.5996	0.6047	0.6074	ESN
crisis n=12	0.5906	0.6023	0.5964	0.6031	HAR-X
calm n=10	0.6244	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.)

HAR-X is best or co-best everywhere; CHIMERA never beats it (slightly worse; raw-significant at n=8/12) and is statistically indistinguishable from the classical ESN/RFF. ****After Holm correction no comparison is significant in either direction.**** The earlier "beats HAR" result was an artifact of comparing against a *feature-poor* HAR: the gain comes from the encoded realized measures (a known SHAR/HARQ effect), not from quantum nonlinearity. ****By our pre-registered criteria, H0 is refuted — we report this negative.**** What honestly survives for the quantum reservoir: it is **competitive** (within ~0.5–1.5% RMSE of the best at every n), **more stable than the recurrent ESN** (lower per-seed variance, e.g. n=12: 0.009 vs 0.015), and it **beats the recurrent ESN on the calm window** (raw p=0.018; n.s. after Holm).

Named canonical baselines (`cli.py run canonical`): running the models a finance referee expects — **SHAR, HAR-CJ, HARQ** (RQ≈RV² proxy) and **HEAVY-RM**, under MSE- *and* QLIKE-loss DM —

none beats HAR-X (confirming it is a fair, strong stand-in, not a strawman), and CHIMERA ties the best of them on RMSE with only a *raw, non-Holm-significant* QLIKE/MZ-efficiency edge (results/canonical_baselines_findings.md).

Rigorous Axis-B: HAR-X (rich linear) is best; quantum shows no significant edge (Holm n.s.)

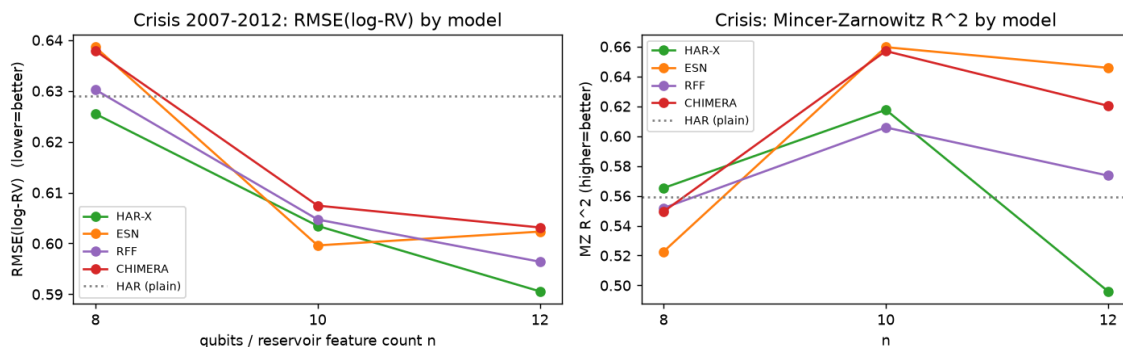


Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n ; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz R^2 .

5.4 Common MNIST benchmark + noise. Same engine, pixels→PCA(n)→ n qubits→Pauli readout→ridge.

Accuracy **scales with qubits** ($n=5/8/10/12$: 0.632/0.798/0.831/0.859, 3 seeds) and ****beats the linear-PCA baseline at every n (real nonlinear lift), confirming sufficient expressivity**** (the benchmark's purpose); a matched ESN **ties or slightly exceeds** CHIMERA (within $\approx 1\%$ for $n \geq 8$; 2.9% at $n=5$) — competitive, not dominant. **Noise:** the classifier is ****invariant to depolarizing**** noise — provably, because depolarizing is a uniform Bloch contraction that per-feature standardization removes exactly (accuracy identical across rates 0.05–0.30) — and **robust to amplitude damping** ($<0.5\%$ at 30%). **Honesty check** (`cli.py run noise_circuit`): a per-Trotter-layer density-matrix study shows the ***converse*** — accumulated two-qubit noise ***during*** evolution is **not** removed by standardization (standardized feature error grows with rate, vs ≈ 0 for readout-only depolarizing), so the invariance above is a property of the ***readout***, not a circuit-level noise-robustness claim; accumulated 2-qubit error over the 380-gate circuit is the real NISQ cost (§6).

5.5 Scaling frontier + quantum-complexity metric. A sparse-exact backend (`expm_multiply`, no dense propagator; matches the dense engine to 2.4×10^4) reaches **$n=16$ exactly**. For the random $\approx 50\%$ -connected reservoir we measure the **bond dimension** χ_{eff} across a balanced cut: it is **full at every n , $\chi_{\text{eff}} = 2^{(n/2)}$** ($16 \rightarrow 256$ for $n=8 \rightarrow 16$) — an exact MPS gets ****zero compression**** — with entanglement entropy $S \approx 1.7\text{--}3.2$ nats (peaking near $n=14$; S is noisier than the rank). The reservoir therefore admits **no low-bond-dimension (MPS/TEBD) shortcut** — exact cost stays exponential (full entanglement is ***necessary, not sufficient*** for true classical hardness) — the precondition any beyond-frontier advantage would need, even though no advantage appears at the simulable scale we can test.

6. Quantum platform and resource planning

Simulator-first on qBraid: dense statevector ≤ 12 qubits, sparse/tensor-network to ≈ 16 , GPU for larger. **Resource estimates** (gate-Trotter, 20 layers; readout is single-basis because all $\mathbf{Z}_i, \mathbf{Z}_j$ are diagonal in the computational basis, so one set of S shots yields all $n+C(n,2)$ observables):

n	two-qubit gates	one-qubit gates	observables	shots S ($\epsilon \approx 1/\sqrt{S}$)	sim wall-clock
8	220	160	36	2k–8k ($\epsilon \approx .011\text{--}.022$)	<1 s/state
10	480	200	55	2k–8k	$\sim$ sec
12	760	240	78	2k–8k	~ 1 min build
16 (sparse)	—	—	136	—	~ 40 min/point

(wall-clock is hardware-dependent — measured on a single CPU here; it is an engineering, not a scientific, quantity.)

QPU validation uses this gate-Trotter circuit on **IonQ / IQM / IBM** via qBraid — the random-sparse Ising needs ***fewer*** two-qubit gates than an all-to-all reservoir, easing NISQ mapping (trapped-ion all-to-all natively supports the arbitrary couplings) — with ****ZNE + measurement mitigation**** and a classical cross-check for every run. The submission path is **executable now** (`qbraid_submit.py, cli.py run qsubmit`): on a simulator it (i) reproduces the engine to 3.9×10^4 via the exact circuit (the classical cross-check we run for every hardware execution), (ii) characterizes the **shot budget** — mean feature error 0.046 / 0.013 / 0.0066 at $S = 256 / 4k / 16k$ shots, the $\epsilon \approx 1/\sqrt{S}$ law, with the gate-Trotter(20) approximation

adding ≈ 0.04 — and (iii) demonstrates **zero-noise extrapolation** recovering toward the noiseless value under a depolarizing sweep. It is one flag (`--device`) from a real backend, pending qBraid credit allocation. We thus characterize all four challenge axes: reservoir size (§5.1/5.5), encoding density (§5.2/5.3), shot budget, and noise (§5.4). *(QCi Dirac-3 is the separate optimization challenge's device.)*

7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the ≤ 16 -qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The S&P 500 RV sample ends Feb-2020 — the 2008 GFC is in-sample, the 2020 COVID shock just outside it; broader assets/periods untested. (iii) Noise is studied on MNIST with single-qubit channels at the readout; full noisy-circuit and shot-noise simulation is deferred to the QPU runs. (iv) g is regularization-dependent (we report the qualitative gap, not a tuned value). (v) **No real-QPU run yet** (simulator cross-checked; pending qBraid credit allocation). (vi) Distinctness and full-rank entanglement are *necessary, not sufficient* for advantage — whether they convert beyond the classical-simulation frontier is the open question.

8. Stakeholder impact, milestone plan, AI disclosure

Reliable volatility forecasts feed portfolio hedging, dynamic risk limits and derivatives pricing. We make the impact **concrete** (`cli.py run economics`): a volatility-timing backtest that sizes S&P-500 exposure by the one-step RV forecast **nearly halves the max drawdown in the 2008 crisis ($-61\% \rightarrow -32\%$) **at a 10% vol target — a deployable risk lever — but a plain HAR**** forecast captures it, and rich features *and* the quantum reservoir add **no** economic value (negative certainty-equivalent fees). So the credible *negative* is decision-useful: the near-term lever is **vol-timing on a simple realized-variance forecast, not quantum hardware. Milestone plan:** (i) pre-register ✓; (ii) scaling + encoding-density sweeps ✓; (iii) adversarial HAR-X/ESN/RFF test ✓; (iv) MNIST + noise ✓; (v) sparse/TN frontier + bond dimension ✓; (vi) gate-Trotter QPU validation (IonQ/IQM/IBM) — simulator cross-checked; fallback = TN + density-matrix noise emulation. **AI disclosure:** Claude (Anthropic) assisted with code and drafting under the team's direction; all formulations, decisions, and results are the team's own, produced by executing team code on public data.

References

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